Motor Control, (Ahead of Print)
https://doi.org/10.1123/mc.2018-0070
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Human Kinetics
ARTICLE

Task Constraints and Coordination Flexibility in Young Swimmers

Ana F. Silva
University of Porto

Pedro Figueiredo
Portuguese Football Federation

Sara Morais, João P. Vilas-Boas, and Ricardo J. Fernandes
University of Porto

Ludovic Seifert
University of Rouen Normandy

This study aimed to examine young swimmers' behavioral flexibility when facing different task constraints, such as swimming speed and stroke frequency. Eighteen (five boys and 13 girls) 13- to 15-year-old swimmers performed a 15 × 50-m front crawl with five trials, at 100%, 90%, and 70% each of their 50 m maximal swimming speed and randomly at 90%, 95%, 100%, 105%, and 110% of their preferred stroke frequency. Seven aerial and six underwater cameras were used to assess kinematics (one cycle), with upper-limb coordination computed through a continuous relative phase and index of coordination. A cluster analysis identified six patterns of coordination used by swimmers when facing various speed and stroke frequency constraints. The patterns' nature and the way the swimmers shifted between them are more important than getting the highest number of patterns (range of repertoire), that is, a change in the motor pattern in order to adapt correctly is more important than being able to execute a great number of patterns.

Keywords: ecological dynamics, motor control, swimming

It is well accepted that a high level of expertise is characterized by motor dexterity, expressing the expert's ability to reach the goal of a task correctly, quickly, reasonably (efficiently), and with resourcefulness (Bernstein, 1996). Resourcefulness (i.e., the initiative and stability of a set of constraints) appears to be the most important property of dexterity, as this latter does not refer to the

Silva, Vilas-Boas, and Fernandes are with the Centre of Research, Education, Innovation and Intervention in Sport (CIFI2D), Faculty of Sport, University of Porto, Porto, Portugal. Silva, Morais, Vilas-Boas, and Fernandes are with LABIOMEP (Porto Biomechanics Laboratory), University of Porto, Porto, Portugal. Figueiredo is with Portugal Football School, Portuguese Football Federation, Lisbon, Portugal. Seifert is with CETAPS EA3832, Faculty of Sport Sciences, University of Rouen Normandy, Mont-Saint-Aignan, France. Silva (anafilsilva@gmail.com) is corresponding author.

movements themselves, but to their adaptation to a set of constraints. In fact, following the ecological dynamics theoretical framework, the successful performer is able to adapt his/her behavior to dynamically shifting environments that characterize sport competition (Seifert, Button, & Davids, 2013), with resourcefulness reflecting his/her capability to functionally vary behavior. Functional variability could occur by changing coordination patterns or by adjusting spatial–temporal parameters without performance outcome deterioration (Seifert et al., 2013). Therefore, although behavior could be characterized by stable and reproducible coordination patterns against perturbations, it is not stereotyped and rigid, but flexible and adaptive (Warren, 2006).

Adaptive behavior does not consist in coordinated movement per se, but in goal-directed environment-orientated action. The ecological dynamics framework postulated for individual–environment system coupling, in which information is viewed as regulating action directly, supports circular causality between perception and action (Davids, Araújo, Hristovski, Passos, & Chow, 2012). Thus, the function of perception and action is to stabilize behavior on the task goal while maintaining adaptive flexibility (i.e., to achieve the task goal the performer should use information to improve adaptive flexibility instead of adopting a rigidly stable solution; Warren, 2006, p. 358). Thus, this adaptive flexibility is related to the ability to adapt structurally (i.e., modification of movement or coordination pattern) to dynamic and interacting constraints without compromising function or yielding the same output (Edelman & Gally, 2001).

Newell (1986) proposed to manipulate three types of constraints, which act as boundaries for the perceptual-motor workspace exploration, instead of using prescriptions: (a) organismic, referring to the structural or functional characteristics of performers; (b) environmental (external to the organism), referring to contextual characteristics; and (c) task, related to the goal of the activity, the rules or instructions, and the devices used. Constraints are used as boundary conditions that both limit or facilitate the action possibilities and the coordination patterns available, interacting in complex configurations, shaping the perceptual-motor workspace of each learner (Davids et al., 2012). Therefore, it has been argued that skill adaptation (rather than its acquisition) can be guided through the manipulation of the key constraints acting on the task (Newell, 1986), that is, applying a constraint-led approach. This approach clearly emphasizes searching strategy and exploration during learning, rather than using verbal instruction and an information-processing perspective (Davids, Araújo, Shuttleworth, & Button, 2003). Regarding motor coordination in cyclic locomotion tasks (e.g., Kelso, 1995) and particularly in swimming (e.g., Kelso, 1995; Seifert, Button, & Brazier, 2010; Seifert, Chollet, & Rouard, 2007), speed and movement frequency appeared as key constraints. In the front crawl, for instance, speed and stroke frequency appear as the main constraints on upper-limb coordination (Potdevin, Bril, Sidney, & Pelayo, 2006; Seifert, Chollet, & Rouard, 2007), with a quadratic regression observed between the increase of speed and the change of the upper-limb coordination (Seifert & Chollet, 2010). Although expert swimmers usually switch from a catch-up to a superposition upper-limb coordination pattern up to a speed of ~1.80 m/s and/or a stroke frequency of ~50 cycles/min (Seifert, Chollet, & Bardy, 2004), interindividual variability occurs, reflecting the individual adaptation of coordination to constraints (Bideault, Herault, & Seifert, 2013). This observation becomes more obvious when

swimmers of various skill levels are assessed; indeed, nonexpert front crawl swimmers changed their upper-limb coordination from the catch-up pattern to the superposition pattern at a stroke frequency between 45 and 50 cycles/min, whereas expert swimmers exhibited this switching between 50 and 55 cycles/min (Potdevin et al., 2006). Moreover, competition analysis showed that stroke frequency can really vary within and between laps (Hellard et al., 2008; Simbaña Escobar, Hellard, Pyne, & Seifert, 2018), inviting us to consider and manipulate speed and stroke frequency, as they showed to be the main constraints to evidence a certain coordination pattern. Nevertheless, training swimmers to perform at different stroke frequencies to enlarge their behavioral flexibility is not a common strategy among coaches, as they often try to stabilize a given stroke frequency that swimmers should keep stable along the event. Thus, the constraint-led approach could be particularly meaningful for young swimmers who have not yet stabilized a certain stroke frequency and upper-limb coordination pattern, notably due to changes in strength and body size with age. In fact, changes in muscle development in young swimmers influence muscle length–tension and force–velocity relationships, which could improve swimmers' performance (Nasirzade et al., 2014). Complementarily, the increase in anthropometrical characteristics during growth has been indicated to be the main reason for performance changes, through its influence in biomechanical parameters (e.g., stroke length increases; Nasirzade et al., 2015). In fact, these modifications are related to the maturation process that tends to begin at age 8–10 years in girls and 10–12 years in boys, reaching its peak of development ~14 years (after the peak high velocity; Malina, Bouchard, & Bar-Or, 2004).

The main aim of the current study was to examine young swimmers' behavioral flexibility (through upper-limb coordination patterns) when task constraints (swimming speed and stroke frequency) are manipulated, investigating the relevance of training around a swimmer's preferred stroke frequency to enlarge his/her behavioral flexibility. We hypothesized that high behavioral flexibility might correspond to (a) a great repertoire of upper-limb coordination patterns, allowing swimmers to switch between coordination patterns, and (b) a high stability of upper-limb coordination pattern within a speed range (slow vs. fast speed) and/or stroke frequency (low vs. high stroke frequency). Complementarily, we examined the interindividual coordination variability to cope with the above-referred task constraints and hypothesized that the swimmers change their upper-limb coordination patterns either with swimming speed, with stroke frequency, or with the combined effect of stroke frequency and swimming speed.

Methods

Participants

Eighteen young swimmers (five boys and 13 girls) participated in this study. Their main characteristics were 14.8 ± 0.4 versus 13.6 ± 0.8 years old, 175.0 ± 7.2 versus 163.2 ± 5.7 cm of height, 177.9 ± 7.3 versus 166.2 ± 5.7 cm of arm span, 68.3 ± 4.1 versus 56.1 ± 6.5 kg of body mass, 9.0 ± 2.2 versus 7.9 ± 3.0 years of practice, and performance over the 50-m freestyle event of 81.7% ± 1.0% versus 78.0% ± 1.5% of the short course world record for boys and girls, respectively. All participants

were in the same skill level, and they incorporated the national competition level. Hence, these swimmers did not receive any specific training focusing on stroke frequency prior to the current study, but they were familiar with the metronome for stroke frequency pacing. An informed consent was signed by the swimmers' parents, and coaches well knew the experimental protocol. The swimmers were informed about the experimental procedures, which were approved by the local ethics committee and performed according to the Declaration of Helsinki.

Experimental Procedure

After a standard warm-up (where swimmers could familiarize themselves with the metronome), each participant performed the 15 × 50-m front crawl, with in-water starts, in a 25-m indoor-heated swimming pool with a depth of 1.90 m without breathing in the center of the pool, to avoid the breathing effect on coordination. Each swimmer performed five 50-m trials at 100%, 90%, and 70% of their 50-m maximal speed; however, a variability of 2.5% around the target time was accepted (regarding the previously calculated individual target value). Those percentages were chosen to better characterize a slow, submaximal, and maximal swimming pace, respectively. Between each 50-m trial, swimmers rested for 5 min, so that fatigue did not influence their performance. Bouts 1–5, 6–10, and 11–15 were performed randomly at 90%, 95%, 100%, 105%, and 110% of their preferred stroke frequency, which was calculated in the middle of the pool, between three consecutive upper-limb cycles, using a stopwatch. To contemplate possible variations during the 50-m swims, stroke frequency was calculated in both 25-m laps of the 50-m bouts (and the average values were used). At each swimming speed, the first repetition aimed to determine the swimmers' preferred stroke frequency; for that reason, the metronome was not used in those trials, and the speed was only monitored through the visual pacer, placed in the bottom of the pool, with a flash every 5 m (Pacer2Swim OEM Kulzer TEC, Aveiro, Portugal). When the swimmers did not perform both the required target speed and the stroke frequency, which were monitored using an underwater metronome placed inside the swim cap near the ear (Tempo Trainer Pro; Finis®, Livermore, CA), the trial was repeated.

Apparatus

To record the swimmers' performance, a 13-camera dual-media motion capture (MoCap) setup was used, with seven land plus six underwater cameras (Oqus 3+ and Oqus Underwater; Qualisys AB, Gothenburg, Sweden) operating at 100 Hz. The calibrated volume was defined using an underwater, above water, and twin system to merge the first and the latter calibrations (according to the manufacturer's guidelines). This enabled the creation of a 3-dimensional dual-media working volume (~5.50 × 2.00 × 1.80 m), where the orthogonal axes were defined as x, y, and z for horizontal, mediolateral, and vertical ($z = 0$ defines the water surface) movements, respectively. Data acquisition was performed with Qualisys Track Manager (version 2.7; Qualisys AB), with swimmers wearing 10 anatomical reflective landmarks on the shoulders, elbows, wrists, midfingers, and hips, enabling automatic frame-by-frame digitization using the Qualisys software.

The data treatment was performed with the same system, by identifying manually the different anatomical points.

Data Processing

Stroking Parameters. Although swimming speed and stroke frequency were controlled during the entire test, they were also calculated after the digitizing process, with the former assessed through the horizontal displacement of the hip during one upper-limb cycle over its total duration and the latter as the result of the inverse of the time needed to complete one upper-limb cycle (defined as two consecutive water entries of the same hand). Stroke length was also obtained by the horizontal displacement of the hip through one upper-limb cycle.

Upper-Limb Coordination Analysis. Coordination between right and left upper limbs was assessed through the continuous relative phase (CRP) that has been considered appropriate for cyclic and sinusoidal signals (Lamb & Stöckl, 2014). Its assessment between the upper limbs (arm–shoulder–trunk angle, using the midfinger, shoulder, and hip markers) was performed for one single cycle during the 50-m front crawl, which was recorded in the central part of the pool to avoid the start, turn, and breathing effects. Cycle duration was expressed in percentage allowing its comparison. As CRP was typically derived from the position–speed phase planes of two predominantly sinusoidal oscillators, it contains spatial and temporal information, enabling the computation of angular velocities from positional data for left and right body sides. Widely used in human movement coordination, phase portraits were computed after normalization (Fuchs, Jirsa, Haken, & Kelso, 1996), with the zero value considered to be arbitrary, that is, without any qualitative meaning (Lamb & Stöckl, 2014), and with the amplitude differences between the oscillating segments not affecting coupling measures (Kurz & Stergiou, 2002). Angular displacements and angular velocities were normalized (θ_{norm} and ω_{norm}, respectively) in the interval [−1, +1] as follows (Kurz & Stergiou, 2002; Lamb & Stöckl, 2014):

$$\theta_{norm} = \frac{2\theta}{\theta_{max} - \theta_{min}} - \frac{\theta_{max} + \theta_{min}}{\theta_{max} - \theta_{min}},$$

where θ_{max} and θ_{min} are the maximum and minimum angular positions within one complete upper-limb cycle, respectively, and

$$\omega_{norm} = \frac{2\omega}{\omega_{max} - \omega_{min}} - \frac{\omega_{max} + \omega_{min}}{\omega_{max} - \omega_{min}},$$

where ω_{max} and ω_{min} are the maximum and minimum angular velocities within one complete upper-limb cycle, respectively.

Although some authors presented the CRP results without any normalization in the phase angle, Kurz and Stergiou (2002) highlighted the differences when conducting normalization or not. Therefore, the following reference phase angle was used:

$$\varnothing = \tan^{-1}\left(\frac{\omega_{norm}}{\theta_{norm}}\right).$$

Continuous relative phase was found through the subtraction of the phase angle between oscillators at each point time over the entire cycle (i.e., the left shoulder phase angles were subtracted from the right ones; Figure 1). The CRP values can range from 0° to 360°, and three different modes could be found: (a) in-phase (0° or 360°, as we are referring to a Cartesian referential), when a synchronization between the upper-limb actions is observed; (b) antiphase (180°), when an opposite action is observed; and (c) out-of-phase, when there is a lag time between the two upper-limb actions. Nevertheless, as a variation of ±30° was accepted for the determination of a coordination pattern (Bardy, Oullier, Bootsma, & Stoffregen, 2002; Seifert, Delignieres, Boulesteix, & Chollet, 2007), an in-phase occurs when CRP varies between 0° ± 30° and 360° ± 30° (i.e., when 330° < CRP < 30°), an antiphase is considered when their parameters range between 180° ± 30° (i.e., when 150° < CRP < 210°), and an out-of-phase is observed when the CRP assumes values outside the above-referred ranges (i.e., when 30° < CRP < 150° and 210° <CRP < 330°).

According to Seifert et al. (2011), different parameters were extracted to examine the coordination between upper limbs and its variability within cycle: (a) the mean CRP and its variability through the *SD* of CRP over a cycle; (b) the relative time spent in in-phase, out-of-phase, and antiphase (all expressed in percentage) to inform about the coupling between upper-limb coordination (and not propulsive action coupling); and (c) the relative time between two propulsive upper-limb actions, corresponding to the index of coordination (IdC; Chollet, Chalies, & Chatard, 2000)—the time between the beginning of propulsion of the first right cycle and the end of propulsion of the first left upper-limb cycle and between the beginning of propulsion of the second left upper-limb cycle and the end of propulsion of the first right upper-limb cycle. IdC was calculated based on the division of the arms' actions in four phases: (a) entry/catch, corresponding to the time from the entry of the hand in the water until it starts to make the backward movement; (b) pull, from the end of the previous action until the vertical alignment of the shoulder is achieved (first propulsive phase); (c) push, from the end of the previous action to the exiting of the hand from the water (second propulsive phase); and (d) recovery, covering the time from the exit of the hand until its new entry.

Figure 1 — Angle assessment in one side of the body (arm–shoulder–trunk angle).

The following three different synchronization modes are possible to identify in front crawl: (a) opposition (IdC = 0%), when one upper limb begins the propulsive phase and the other finishing it, providing continuous motor actions; (b) catch-up (IdC < 0%), when a lag time exists between the propulsive phases of the two upper limbs; and (c) superposition (IdC > 0%), describing an overlap in the propulsive phases of both upper limbs.

The rationale used was based on our first goal, which aimed to identify the nature of the coordination by using mean CRP over the cycle. However, as already mentioned by Figueiredo, Seifert, Vilas-Boas, and Fernandes (2012), we expect that the swimmers did not exhibit the same upper-limb coupling all along the cycle. Therefore, *SD* is the second indicator that can indicate any fluctuation of relative phase. Then, to go deeper into the nature of the upper-limb coupling, we have quantified the time spent in each pattern of coordination (in-phase, antiphase, and out-of-phase). Finally, those previous indicators did not inform only on the coupling between the propulsive actions, but also on the coupling between joints; therefore, our second goal was to include IdC (Chollet et al., 2000).

Statistical Analysis

Assuming that swimmers can individually adapt to the various speed and stroke frequency conditions, we did not perform a group comparison based on average data. Instead, we performed a cluster analysis to identify the presence of potential groupings within the whole set of individuals regarding swimming speed and stroke frequency. It enables one to detect individual coordination profiles and to assess the upper-limb coordination dynamics, that is, the time distribution of the coordination patterns. Cluster analysis has been widely used to examine movement and coordination pattern variability within and between individuals, as no a priori assumptions regarding data set structure (e.g., the normality of the distribution) are required (Duda, Hart, & Stork, 2001). Furthermore, as this process is an unsupervised machine learning (Duda et al., 2001), it organizes groups without human intervention and inherent bias by identifying homogeneous individual clusters that share several common characteristics. In our study, clustering was performed on six kinematic parameters, each of them relating to upper-limb coordination (mean CRP; *SD* of CRP; time spent in in-phase, in out-of-phase, and in antiphase; and IdC) for each bout, so that natural groupings could be observed.

The Bayesian information criterion index was used as a model selection criterion to validate the number of clusters found within the data set. It was performed for 2–12 clusters and represented in a vector containing the above-referred index for the number of clusters, corresponding to the first local maximum that best fitted the data set (i.e., the number representing the highest ratio between inter- and intracluster distances; Ludden, Beal, & Sheiner, 1994). The cluster analysis ran once for the entire data sets, comprising 270 trials (18 swimmers × 3 speeds × 5 conditions of stroke frequency) using the Fisher-EM algorithm (Bouveyron & Brunet-Saumard, 2014), based on a probabilistic model that projects the data in a latent subspace at each iteration in such a way that emerging clusters maximize the Fisher information (Seifert et al., 2016). All statistical procedures were performed with RStudio© (0.99.491, 2009–2015; RStudio, Inc, Boston, MA).

Results

Behavioral Profile According to Speed and Stroke Frequency

The sparsity values were 0.81 for CRP, 0.98 for *SD* of CRP, 0.97 for the time spent in in-phase, 0.92 for the time spent in out-of-phase, 0.98 for the time spent in antiphase, and 0.88 for IdC, suggesting that these six parameters were informative to cluster our 270 observations. Six different clusters were distinguished as follows (Table 1): (a) cluster 1 (composed by 16 observations) displayed a slower speed, stroke frequency, and IdC, with a greater relative time in the in-phase pattern; (b) cluster 2 (included 99 observations) showed intermediate speed and IdC values and lower *SD* of CRP, although a high relative time in the antiphase pattern; (c) cluster 3 (with 17 observations), although showing a slower value of speed and more 50-m trials with the second highest value of stroke frequency, displayed the second highest value of IdC, with lower CRP and time percentage in antiphase (as this cluster presented the highest value of out-of-phase); (d) cluster 4 (with only five observations) showed high CRP, *SD* of CRP, and relative time spent in the in-phase pattern, although presenting the lowest value of IdC; (e) cluster 5 (with 107 observations) showed the highest speed, stroke frequency, and IdC values of all clusters, but a lower relative time spent in in-phase; and (f) cluster 6 (with 26 observations) expressed an intermediate speed value, with a quite low stroke frequency and with intermediate values of CRP and *SD* of CRP compared with other clusters.

Table 2 shows the clusters' distribution considering each speed and stroke frequency condition without considering interindividual variability.

Interindividual Behavioral Variability

Figure 2 shows the cluster distribution considering each condition of speed and stroke frequency for each swimmer. The majority of young swimmers ($n = 8$) switched between four clusters in the different conditions imposed by the 15×50-m test. It was also observed that switching three and five times between clusters was the second most used profile ($n = 3$ each). Only two swimmers switched between the six clusters and between the two clusters. It was also noted that swimmers with greater speed and stroke frequency variation between trials switched between clusters 3 and 6.

Discussion

Behavioral Profile According to Speed and Stroke Frequency

Cluster analysis showed different behavioral dynamics when speed and stroke frequency were manipulated, confirming that these two parameters seem to act as the main constraints influencing the system behavior (Potdevin et al., 2006). Indeed, although a catch-up mode of coordination between the propulsive phases of the upper limbs was always observed, six different clusters emerged through the various ranges of speeds and stroke frequencies imposed by the experimental protocol. The existence of the catch-up coordination mode, even at the

Table 1 ***M*** **±** ***SD*** **Values of Speed, Stroke Frequency, CRP,** ***SD*** **of CRP, In-Phase, Antiphase, Out-of-Phase, and IdC of Each Cluster**

	Cluster 1 (*n* = 16)	Cluster 2 (*n* = 99)	Cluster 3 (*n* = 17)	Cluster 4 (*n* = 5)	Cluster 5 (*n* = 107)	Cluster 6 (*n* = 26)
Speed (m/s)	1.09 ± 0.14	1.25 ± 0.20	1.14 ± 0.12	1.21 ± 0.21	1.42 ± 0.17	1.26 ± 0.24
Stroke frequency (cycle/min)	34.3 ± 8.5	38.1 ± 8.0	41.7 ± 5.9	34.2 ± 9.1	44.0 ± 8.7	36.8 ± 11.5
CRP (deg)	226.4 ± 10.0	195.4 ± 12.5	183.0 ± 16.8	256.0 ± 25.2	190.3 ± 12.0	227.0 ± 9.8
SD of CRP (deg)	74.8 ± 11.1	44.2 ± 11.6	68.8 ± 15.0	115.8 ± 21.9	47.5 ± 7.4	66.2 ± 11.8
In-phase (%)	6.9 ± 4.9	1.7 ± 1.9	1.8 ± 1.7	5.6 ± 5.3	0.8 ± 0.8	4.9 ± 2.9
Antiphase (%)	63.7 ± 7.9	64.2 ± 7.5	22.6 ± 7.0	41.7 ± 8.4	43.6 ± 7.1	39.4 ± 7.1
Out-of-phase (%)	29.4 ± 4.9	34.1 ± 7.1	75.6 ± 7.8	52.7 ± 8.9	55.7 ± 7.2	55.7 ± 7.4
IdC (%)	−12.0 ± 7.3	−10.2 ± 5.3	−9.6 ± 4.5	−14.5 ± 4.3	−9.1 ± 5.1	−12.0 ± 5.7

Note. CRP = continuous relative phase; IdC = index of coordination.

Table 2 Percentage of Occurrence of the Different Clusters for Each Swimmer Within Stroke Frequency Condition

Speed (%)	Stroke frequency (%)	Cluster 1 ($n = 16$)	Cluster 2 ($n = 99$)	Cluster 3 ($n = 17$)	Cluster 4 ($n = 5$)	Cluster 5 ($n = 107$)	Cluster 6 ($n = 26$)
	90	0	5.1	11.8	0	10.4	0
	95	0	5.1	11.8	0	10.4	0
100	100	0	4.0	17.6	0	9.3	3.8
	105	0	2.0	5.9	40	9.3	11.6
	110	0	7.1	5.9	0	8.4	3.8
	90	0	6.0	17.6	0	6.5	7.7
	95	0	12.1	0	0	4.7	3.8
90	100	6.3	2.0	17.6	0	9.3	7.7
	105	0	3.0	11.8	0	11.2	3.8
	110	0	10.1	0	0	7.5	0
	90	25	8.1	0	0	2.8	11.6
	95	12.5	12.1	0	0	0.9	11.6
70	100	12.5	6.1	0	20	3.7	19.2
	105	12.5	9.1	0	40	1.9	11.6
	110	31.2	8.1	0	0	3.7	3.8

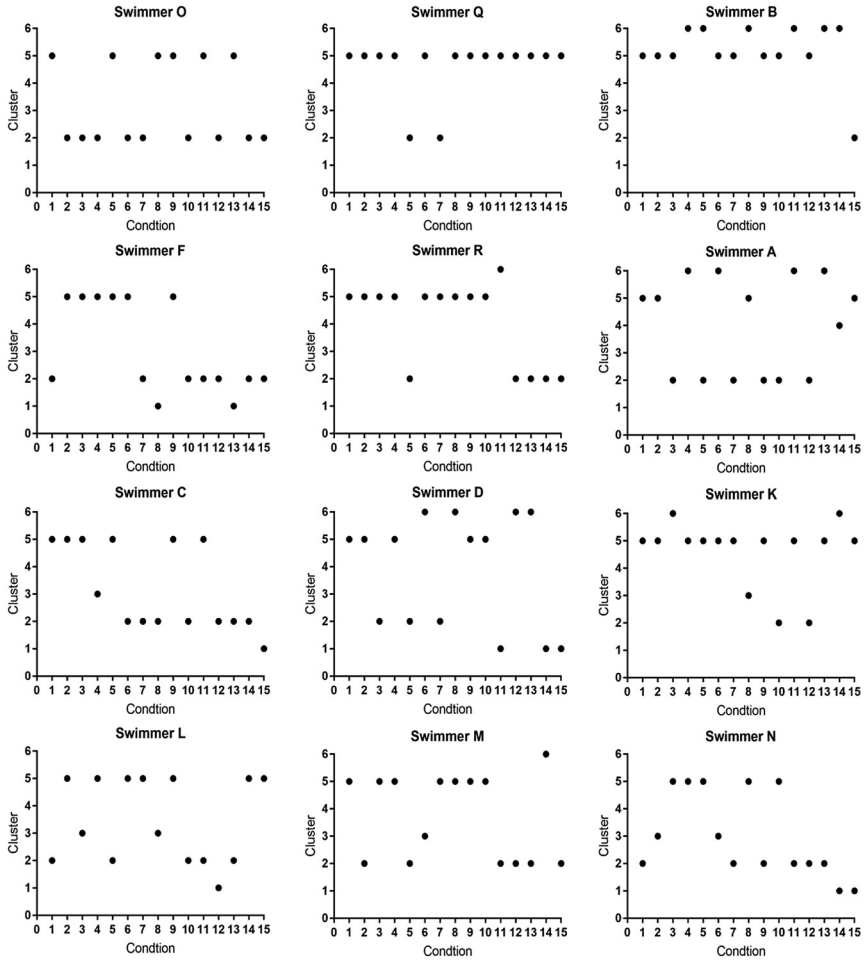

Figure 2 — Clusters distribution through the 15 different conditions (with sequence presented on Table 2, starting with speed 100% and 90% of preferred stroke frequency and finishing with speed 70% and 110% of preferred stroke frequency) displayed by swimmers with: two clusters adopted (O and Q), three (B, F, and R), four (A, C, D, K, L, M, N, and P), five (E, I, and J), and six (G and H).

fastest speeds, was in accordance with the literature for young swimmers (e.g., Figueiredo, Silva, Sampaio, Vilas-Boas, & Fernandes, 2015; Silva et al., 2012). Our main findings (summarized in Tables 1 and 2) highlighted that clusters 1 and 5 seemed to be functional behaviors and complemented each other, as cluster 1 was mainly used at lower speeds, while cluster 5 was mostly adopted for the maximal and submaximal (100% and 90% speed, respectively) conditions, whatever the stroke frequency. The four other clusters represent mixed strategies, combining speed and stroke frequency differently.

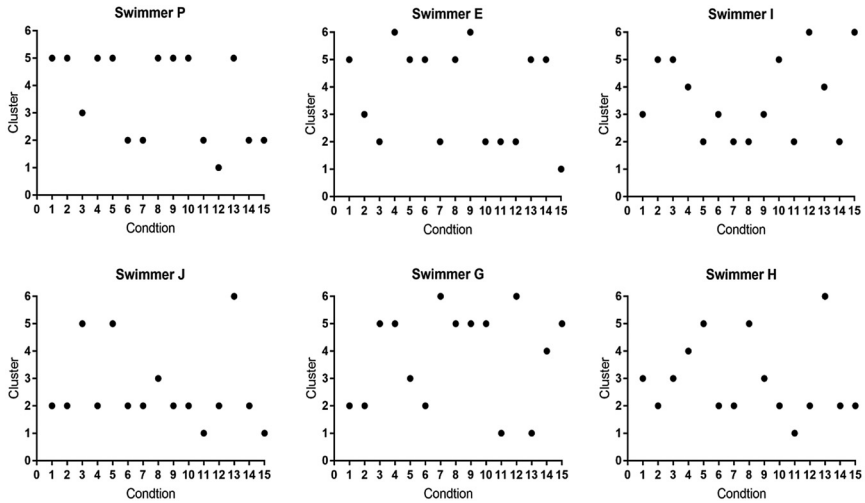

Figure 2 — (continued)

Cluster 1 included 50-m bouts mainly performed at 70% maximal speed, where the stroke frequency seemed to have not exerted a preponderant effect, as it contained a similar number of 50 m repetitions at different percentages of stroke frequencies (confirmed in the tables). Indeed, in this cluster, the speed (relative and absolute) and stroke frequency presented the lowest values. Consequently, a lower IdC, with the highest value of relative time in the in-phase pattern, was exhibited, which is appropriate for slow paces, as it favors the glide phase, mainly for buoyancy reasons (Seifert et al., 2004). Indeed, although swimming locomotion results in both the ability to produce propulsive forces and to minimize the resistive ones (Toussaint & Truijens, 2005), it is known that swimmers favor propulsion generation at sprint paces, giving great importance to active drag minimization at the slowest paces (McCabe & Sanders, 2012). This resistance minimization includes a great glide phase, where swimmers search for an efficient hydrodynamic position, confirming the catch-up mode of coordination.

Cluster 5 represented the most frequent pattern adopted by swimmers, with the speed and stroke frequency effect appearing to be combined, as all stroke frequency conditions were present in the trials in which the swimmers swam the 50 m at 90% and 100% of their maximal speed. Regarding CRP values, the out-of-phase coupling was the most dominant pattern (~55% of the cycle duration), and considering that front crawl is characterized as an alternated swimming technique, a greater antiphase mode would be expected. Nevertheless, the transitions between the aerial (arm recovery) and underwater (arm catch, pull, and push) phases explain that the CRP values are not only in antiphase, but also more or less in out-of-phase coupling. This cluster also exhibited the highest IdC value (although remaining negative), expressing a catch-up coordination pattern. As ages between 11 and 14 years old are characterized as a period of versatility for mastering the technique and preparing for a progressive workout (Sozański, 1999), it could explain

why young swimmers could not achieve the same upper-limb coordination pattern as adult elite swimmers, who adopt opposition or even a superposition coordination pattern when performing at maximal speeds (e.g., Chollet et al., 2000).

As suggested previously, the four other clusters showed mixed strategies with various combinations of speed and stroke frequency conditions. For instance, cluster 3 included trials swum at maximal and submaximal speeds (100% and 90%) and low stroke frequencies (varying between 90% and 100%). However, when compared with other clusters, this did not show the highest speed values, presenting the second lowest speed and the second highest stroke frequency values (Table 1). Moreover, regarding the absolute speed value, it was expected that very negative IdC values would be displayed, but compared with other clusters, it presented the second highest IdC value. Therefore, although this cluster expressed a lower speed and stroke frequency (in absolute) compared with others, its IdC value remained high (the second highest), showing that the combination of high speed and stroke frequency could play an important role. In the literature, positive correlations between IdC with speed (e.g., $r = .37$, Lerda & Cardelli, 2003; $r = .69$, Seifert et al., 2004) and stroke frequency (e.g., $r = .54$, Chollet et al., 2000; $r = .76$, Seifert et al., 2004) were already observed. Another good example of a mixed strategy was observed in cluster 4 (the one with the lowest number of trials), which was mainly characterized by lower speed conditions (Table 2), but with intermediate absolute speed values (Table 1). Moreover, a great percentage of the cycle was in the out-of-phase pattern, expressed by a CRP value, and a very high *SD* of CRP was reported, suggesting an unstable irregular coordination pattern.

The second most used coordination pattern was the one represented by cluster 2, which expressed an intermediate speed and stroke frequency values, and the longest time spent in antiphase coupling. This coordination pattern seemed to be very consistent, as this cluster presented the lowest *SD* of CRP value. All these characteristics seem to suggest that coaches do not often manipulate task constraints in young swimmers' training sessions, as a great number of trials were in this cluster, and they corresponded to various conditions of speed and stroke frequency. Therefore, the development of functional flexibility (Warren, 2006) might be achieved by training young swimmers to explore various coordination patterns when they swim at different stroke frequencies and swimming speeds.

Finally, cluster 6 exhibited a great number of trials swum at 70% and 90% of maximal speed, with various conditions of stroke frequency, except the 110% of preferred stroke frequency. Surprisingly, this cluster was also used for fast swim speed conditions (100% of the maximal speed) for high stroke frequency conditions (100%–110% of preferred stroke frequency). Furthermore, when observing CRP values and the *SD*, as well as the IdC and the relative time spent in each coupling pattern, it showed balanced values compared with other clusters, probably because this pattern was not used by the same swimmers for the same conditions. In fact, although some swimmers adopted the cluster 6 for slow speed and low stroke frequency, some other swimmers used it for high speed and high stroke frequency; this pattern well exemplifies interindividual behavioral variability in stroke frequency and speed management.

Interindividual Behavioral Variability

In the current study, some young swimmers exhibited a larger coordination repertoire and switched between several clusters, whereas some others remained in few clusters. Moreover, some swimmers switched very often, showing an inconsistency in their adaptation to task constraints, while some others exhibited more stability according to swimming speeds and/or stroke frequency. Together, these two findings suggest that interindividual behavioral variability could be adaptive flexibility, as previously hypothesized. In fact, not all variability has shown to be adaptive, as our study proposed that the patterns nature and shifting appropriately between them (according to speed and stroke frequency) seem more important than getting high behavioral changes (i.e., high number of switching) or/and a high range of the repertoire (i.e., high number of patterns). Indeed, a functional adaptation could be accepted when an optimal relationship between behavioral pattern and performance outcome occurs. Thus, in the current study, adaptive flexibility would be optimal when the upper-limb coordination pattern showed a link between swimming speed and stroke frequency ranges.

Movement variability is known to follow the central nervous system development, being observed as a U-shape function to characterize the relationship between intraindividual variability in performance and age across the life span, showing a decrease through childhood and adolescence to adulthood (MacDonald, Nyberg, & Bäckman, 2006). Following this, one hypothesis could be that the motor variability observed in our study might relate to different stages of development in the central nervous system. However, the swimmers included in the current study are postpubertal; therefore, the central nervous system has already reached its maturation, and we hypothesized that the motor variability could be related to sensory motor system organization. In fact, Deutsch and Newell (2001) showed that motor performance improvement with age further relates to a better ability to organize the sensory motor system to match the task demands rather than reductions in the system noise.

According to the previous description provided for cluster 5, the swimmers were considered to functionally adapt their behavior when using this pattern in the fastest speed and highest stroke frequencies. However, some swimmers (O, K, L, and E) have used that cluster in the lowest speed, denoting less appropriate behavior, as it exhibited an upper-limb coordination pattern further adapted for fast speeds and high stroke frequencies. Likewise, when swimmers used clusters 3 (C, K, L, N, P, E, I, G, and H) and 4 (A, I, G, and H), they exhibited a less appropriate behavior, showing mixed strategies when compared with other clusters, considering the requested condition and the obtained result (comparison between tables). Finally, clusters 2 and 6 seemed to present "intermediate" states, as swimmers exhibited an intermediate IdC value compared with other clusters, and these clusters were used in contrasted conditions of speed and stroke frequency. These intermediate states can be viewed as inappropriate and nonadaptive, but they can also reflect a transient behavior used to search for new motor solutions and to explore a new coordination pattern.

We hypothesized that behavioral fluctuations can be crucial in the training process, helping individuals to search for more varied and functional coordination solutions to fit the task dynamics. Davids et al. (2012) stated that there is a limited

number of varied but stable performance solutions that can be achieved for a desired outcome. Indeed, and corroborating with those authors, swimmers Q, R, and D, who switched among clusters 2, 3, and 4, seemed to show an adaptive flexibility, as they were included in cluster 5 in the maximal and submaximal speeds and cluster 1 in the slow speed (70% speed). Thus, they might use the "intermediate" clusters (2 and 6) in some conditions to explore new possibilities. Hence, these results confirmed our first hypothesis, as these swimmers clearly switched between clusters with various coordination patterns, showing not only a large repertoire of upper-limb coordination but also an adaptive behavioral flexibility.

Moreover, regarding our second hypothesis, it seemed that adaptive behavioral flexibility further related to coordination switching with speed condition changes than with stroke frequency condition changes. Indeed, we observed a high stability of upper-limb coordination patterns whatever the stroke frequency conditions of a given speed, but more coordination switching between speed conditions. Therefore, manipulating speed conditions could encourage the exploration of different behaviors, notably by designing situations where swimmers can experience a supramaximal swimming speed (such as towing swimmers with an elastic, using a swimming flume, or using a run-and-swim situation). Indeed, it would lead swimmers to stabilize functional individual solutions (Davids et al., 2012).

Study Limitation

The major limitation of this study was the fact that only one upper-limb cycle was analyzed in each 50-m bout. However, as the shortest swim race was analyzed, intercycle variability might be reduced, and in our study, the behavior is less prone to fatigue. One suggestion could be to use an inertial measurement unit to investigate intercycle variability, with high caution to compute the interlimb angle according to recent studies (Guignard et al., 2017; Guignard, Rouard, Chollet, & Seifert, 2017).

Conclusion

When manipulating swimming speed and stroke frequency with dominance for speed, different front crawl upper-limb coordination patterns could emerge, confirming that these parameters act as the main constraints of the system. However, young swimmers did not change their upper-limb coordination, as they remained in catch-up mode, whatever the performed swimming speed and stroke frequency conditions. This means that the explanation of behavioral flexibility must be investigated somewhere other than the nature of the coordination pattern, such as the coordination dynamics. On one hand, our study highlighted that coordination variability might play an important role during training because it can reflect adaptive flexibility to task constraints. On the other hand, our findings showed that coordination variability did not always correspond to functional adaptation, but it is necessary to explore new possibilities. Thus, the present study suggests that the nature of the coordination pattern, switching

appropriately between coordination patterns (according to speed and stroke frequency), seems more important than getting the highest number of switches or the highest number of patterns (range of the repertoire). To achieve that, coaches should use a constraint-led approach to manipulate the constraints, which will allow the swimmers to extend their motor repertoire, giving them opportunities to choose the most adaptive patterns in each condition. These findings show the interest of studying motor variability in the ecological context of performance, which goes beyond the swimming context. Finally, the originality of our study involved investigating motor variability both within and between individuals, notably by studying the effect of age (not by comparing young with adult swimmers, but by examining the individual repertoire of young swimmers).

Acknowledgments

This work was supported by the Portuguese Science and Technology Foundation under grant SFRH/BD/87780/2012.

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